What Do Personalised Reward Models Learn About New Raters? A Systematic Test and a Three-Parameter Baseline

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Abstract

Personalised and pluralistic reward models promise to represent how individual people judge model outputs — through demographic groups, learned user latents, or examples placed in context. We test what such models actually learn about raters they have never seen, under one protocol on three public multi-rater datasets (MultiPref pairwise preferences, DICES conversational-safety ratings, PRISM response ratings): new raters, k of their own ratings for adaptation, evaluation on held-out items. As a reference we use a deliberately simple baseline from measurement theory: a few text-independent offsets per rater on a distributional reward model's output, describing how the rater uses the scale (three numbers on two-sided preferences: how often they say "clearly", "slightly", "tie"). A learner-free analysis first shows where the predictable structure sits: beyond item location, rater-side scale use improves held-out likelihood about 25 times more than item-specific dispersion, while the item-level shape that replicates across rater panels is small and only partly decodable from text. Under the common protocol the scale-use baseline improves held-out log-likelihood by 0.08–0.10 nats per rating at k = 20 on all three datasets; demographic or crowd/expert groups add at most 0.003; a tuned low-rank text-interacting latent matches but does not exceed it; reimplementations of two published personalised reward models (VPL, PAL) gain less, and their user representations do not encode scale use; an in-context-personalised LoRA reward model gains less (two folds); and the gain is unchanged on top of a fine-tuned reward model. Two properties limit what this personalisation buys: on binary preference labels — the format most personalisation methods are trained and evaluated on — every method's gain is at most 0.002 nats, and the gain is mostly calibration (how strongly a rater prefers) rather than ranking (which response they prefer; +0.5 points of direction accuracy on MultiPref). We recommend graded labels, a scale-use baseline, new-rater evaluation, and separate calibration and ranking metrics for personalised reward modelling.

1. Introduction

Reward models and LLM judges are trained on human ratings that disagree. A recent line of work treats this disagreement as signal to be modelled rather than noise to be averaged: distributional preference learning estimates a distribution over each alternative's utility rather than a point score (Siththaranjan et al., 2024); jury learning predicts individual annotators from content, annotator and demographic-group embeddings (Gordon et al., 2022), and large datasets such as PRISM are analysed by sociodemographic group (Kirk et al., 2024); personalised reward models learn a per-user latent, inferred for a new user from a few examples (Poddar et al., 2024; Chen et al., 2025). The pluralistic-alignment agenda (Sorensen et al., 2024) makes the representation of individual and group differences a first-class goal.

What these models learn about a person is rarely tested directly. Evaluations typically use synthetic users whose preferences differ by construction, binary preference labels, or users seen in training. We ask a simple question under a common protocol on real raters: **given k ratings from someone the model has never seen, how much better can it predict their other ratings, and what is it that it learns?**

Measurement theory suggests a candidate answer that reward modelling largely ignores: raters differ in **how they use the response scale** — some say "clearly better" where others say "slightly better", some rarely call a tie, some are harsher across the board (Linacre, 1989; Baumgartner & Steenkamp, 2001). Such response styles are text-independent, cheap to estimate, and invisible to models that condition only on content or demographics. We make them the reference point: if a personalised reward model cannot

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beat a few scale-use offsets per rater, it has not shown that it learns more than how people use the scale.

Contributions. (1) A learner-free measurement of where the predictable structure of multi-rater preference data sits — split-half reliability of the rating distribution's shape, cross-fitted text decodability, and a rater–item decomposition of held-out likelihood (§3). (2) A three-parameter scale-use baseline that plugs into any reward model with graded outputs (§4). (3) A systematic new-rater test, on three datasets, of group-conditioned, low-rank-latent, published (VPL, PAL), in-context and fine-tuned personalised reward models against that baseline, with hypotheses fixed in timestamped internal protocols, post-result changes marked as amendments, and failures reported (§5, §8). (4) Two limits of what personalisation buys — it disappears on binary labels and it is mostly calibration rather than ranking — and recommendations for the field (§6–7).

2. Related work

Pluralistic and distributional reward models. Distributional preference learning (Siththaranjan et al., 2024), diverging-preference detection (Zhang et al., 2025), PRISM (Kirk et al., 2024), MultiPref (Miranda et al., 2025) and DICES (Aroyo et al., 2023) motivate modelling disagreement; jury learning (Gordon et al., 2022) and multi-annotator models (Mostafazadeh Davani et al., 2022) predict individual annotators. **Personalised reward models** infer a user latent from a few examples (Poddar et al., 2024; Chen et al., 2025) or learn per-user embeddings jointly with the reward model (Li et al., 2024). **Annotator demographics** as predictors of labels have been found weak in several studies (e.g. Orlikowski et al., 2023). **Measurement theory**: rater severity and scale use are standard facets in many-facet Rasch measurement (Linacre, 1989; Eckes, 2011) and response styles are a classical survey-methods topic (Baumgartner & Steenkamp, 2001). We connect the last line to the first: for new-rater prediction, the measurement-theory rater facet is what the reward model is missing.

3. Where is the predictable structure? A learner-free measurement

Data. MultiPref (Miranda et al., 2025; 10,461 comparisons; 4 ratings each by 2 crowd and 2 expert raters; 227 persistent raters; 5 ordered levels A-clearly … B-clearly). PRISM (Kirk et al., 2024; 68,371 responses each rated 1–100 by one of 1,396 users; binned to 5 fixed levels). DICES-350 (Aroyo et al., 2023; 350 conversations fully crossed with 104 quality-checked raters; safe / unsure / unsafe).

Estimators. For an item with ratings on K ordered levels, let h be the empirical rating distribution and C the cumulative operator. Split the item's raters at random into disjoint panels P_1 (a raters) and P_2, and let q be a cross-fitted *location-only* forecast with exactly the mode and mean of P_1 (built from other items in the same mode-and-mean cell). With $r_1 = C(h_1 - q)$, $r_2 = C(h_2 - q)$, $M = \|r_1\|^2/n$ and $X = \langle r_1, r_2\rangle/n$, the forecasting value of the raw distribution over its own location obeys the identity $S(h_1) - S(q) = M - 2X$ under the ranked probability score, so $\rho = X/M$ is a split-half reliability of the distribution's *shape* and the raw distribution beats its location exactly when $\rho > 1/2$. Text decodability is the proper-score gain $G = (2\langle \hat{r}, r_2\rangle - \|\hat{r}\|^2)/n$ of a residual $\hat{r}$ decoded from frozen text features, cross-fitted over prompt groups, reported as the fraction G/X of the reproducible shape. The rater–item decomposition fits ordered-probit models $Y^*_{ir} = \theta_i + \delta_r + \sigma_i\sigma_r\varepsilon$ with item locations θ_i, rater shifts δ_r, rater scales σ_r and item dispersions σ_i, compares held-out likelihood over random cells (every held-out rater and item appears in training), and chooses all penalties by inner cross-validation.

Reproducibility. With the split-half estimator of a companion study (anonymised), the reliability of the rating distribution's shape beyond its exact mode and mean is $\rho = 0.25$ [0.22, 0.29] on MultiPref overall preference with three raters (helpful 0.29, harmless 0.21): the raw few-rater distribution forecasts a new rater worse than its own location ($\rho < \frac{1}{2}$). **Text decodability.** A cross-fitted ridge decoder on frozen Qwen2.5-7B-Instruct features, scored on an independent rater panel, recovers G/X = 10.5 % [5.7 %, 11.9 %] of the reproducible shape (within-cell permutation control ≤ 0; positive control 0.35) — comparable to the upper end for ordinal quality ratings (upper bounds 0.4–10 %) and below NLI (17–26 %). **Rater vs item.** A long-format ordered-probit decomposition with real rater identities and penalties tuned by inner cross-validation: rater-specific thresholds improve held-out likelihood by +0.036 nats per rating over item location alone, item-specific dispersion by a further +0.0014 (ratio 25; 19–36 across the five folds); of the rater part, rater scale contributes +0.031 and rater shift +0.005.

Beyond item location, the predictable structure is therefore mostly rater-side scale use, while the item-level shape that replicates across disjoint panels is small and only partly in the text. These two facts come from different estimands and do not say that disagreement is "not in the text". The stage-1 decision rule we had written down returned **"mixed"**, because G/X was not ≤ 10 % (interval 5.7–11.9 %); the factorised model of §4 was chosen with that outcome known.

4. A three-parameter scale-use baseline

Let $\ell(x) \in R\hat{}K$ be the class logits of a text-only distributional reward model for item x (K ordered rating levels). For rater r we add a text-independent offset vector: p(y | x, r) = softmax(ℓ(x) + b_r). Variants: **DF** (b_r $\in R\hat{}K$, free), **DFs**

(two-sided preference scales only: b_r tied symmetrically, (e, s, t, s, e) — extreme, slight, tie — 3 parameters, no A-vs-B position term). Rater offsets have a Gaussian prior centred on the training-rater mean; a new rater's offsets are the MAP estimate from k of their ratings with the text model frozen. Baselines under the same protocol: **D** (text-only distributional RM), **DG** (group offsets: crowd/expert, race, region), **DL** (low-rank text-interacting latent: $\ell(x) + (Wx + w_0)\cdot z_r$, d = 8, the personalised-RM family). An interpretable probit family (rater shift / log-scale on an ordered-probit text head) is reported as a secondary analysis.

Training and adaptation. The text model and the training raters' offsets are fitted jointly by penalised maximum likelihood,

$$\min \sum_{(i,r,y)} -\log p(y \mid x_i, r) + \lambda_{\text{text}}\|V\|^2 + \lambda_r \sum_r \|b_r\|^2,$$

with λ_{text} chosen on the text-only model and λ_r on held-out items of the training raters. A new rater r^* is adapted from k ratings $\{(x_j, y_j)\}$ by $\hat{b}_{r^*} = \arg\min_b \sum_j -\log \text{softmax}(\ell(x_j)+b)_{y_j} + \lambda_r\|b-\bar{b}\|^2$, with $\bar{b}$ the training-rater mean (the prior mean; $\hat{b} = \bar{b}$ at $k = 0$) and the text model frozen — a convex problem in 3–5 parameters, solved in milliseconds. Any text model that outputs graded log-probabilities can be used: we use a ridge head on frozen LLM features (§5.1–5.3), a LoRA-fine-tuned LLM reward model (§5.4) and the decoder of a published personalised reward model (§5.5).

5. Experiments

Protocol. 5 folds; in each, 20 % of raters (stratified by group) and 20 % of prompt groups are held out; models are trained on training raters × training items with penalties chosen on a validation split; each test rater is adapted on k ∈ {0, 5, 10, 20, 50} of their ratings of *training* items (5 random draws) and evaluated on their ratings of *held-out* items. Primary metric: held-out log-likelihood per rating; also RPS and preference-direction accuracy; rater-cluster bootstrap (B = 2,000). Text model in §5.1–5.3: frozen Qwen2.5-7B features with a ridge head; §5.4: a LoRA-fine-tuned reward model.

5.1. Main result (MultiPref)

Figure 1 shows the gain over the text-only reward model as a function of k on all three datasets. In the table, DF/DFs/DG use the training-rater prior and DL and the DFs − DL column use the few-shot-tuned prior (the DFs − DL contrast is post hoc).

Preference-direction accuracy (D: 0.674): DFs +0.005 [+0.001, +0.008] at k = 20; DF −0.014 [−0.020, −0.007]. A three-number style model gains as much as a tuned 8-dimensional text-interacting latent (tie at k = 20) and more at k = 5, 10 and 50; fitting a rater's A-vs-B position from few ratings costs direction accuracy.

5.2. Replications (k = 20)

DF − D and DG − D use the training-rater prior, DF − DL the few-shot-tuned prior. Direction accuracy on PRISM means good-vs-bad around the scale midpoint.

5.3. What the offsets encode

In the probit family, rater *scale* matters more than rater *shift* on MultiPref (+0.036 [+0.020, +0.057] at k = 20), but rater *shift* dominates on DICES (−0.053) and PRISM (−0.031): leniency/strictness on single-response rating scales, intensity on two-sided preference scales.

5.4. On a fine-tuned reward model

With a LoRA-fine-tuned Qwen2.5-7B reward model (D-LoRA, trained per fold on the training raters' rating distributions) as the text model, the three-number style model still adds +0.100 [+0.071, +0.130] nats per rating at k = 20 — the same as its gain on the frozen-feature head (+0.100). Fine-tuning improves the text-only model by +0.027 [+0.016, +0.039] (direction accuracy +1.4 points) and the style model by the same +0.027: the two gains add. Personalising the frozen-feature model with 20 ratings per rater gains more than fine-tuning the text model (+0.072 [+0.041, +0.106]). DFs keeps the direction accuracy of D-LoRA (+0.002 [−0.002, +0.006]). Against in-context personalisation — a separate LoRA model on the same base, trained with ten of the rater's other ratings in the prompt and evaluated with the new rater's k ratings in the prompt (two folds, one training seed per fold, 86–90 raters; the in-context gain varies across folds, +0.075 and +0.021) — the three offsets on D-LoRA are better at k = 5 (+0.041 [+0.016, +0.065] nats per rating) and at k = 20 (+0.042 [+0.020, +0.065]), and more accurate on which response the rater prefers (+2.9 points); the in-context model learns from the history (+0.048 over D-LoRA) but loses 3.0 points of direction accuracy, at several thousand prompt tokens per prediction.

5.5. Published personalised reward models

We reimplemented VPL (Poddar et al., 2024) and PAL-B (Chen et al., 2025) from their official code as heads on the same frozen per-response embeddings and evaluated them under the same new-rater protocol, refitting the text-only and style models on those embeddings. On graded labels, at k = 20 the style model is better than VPL by +0.092 [+0.068, +0.119] nats per rating and than PAL-B by +0.158. VPL's latent does carry rater information once its posterior collapse

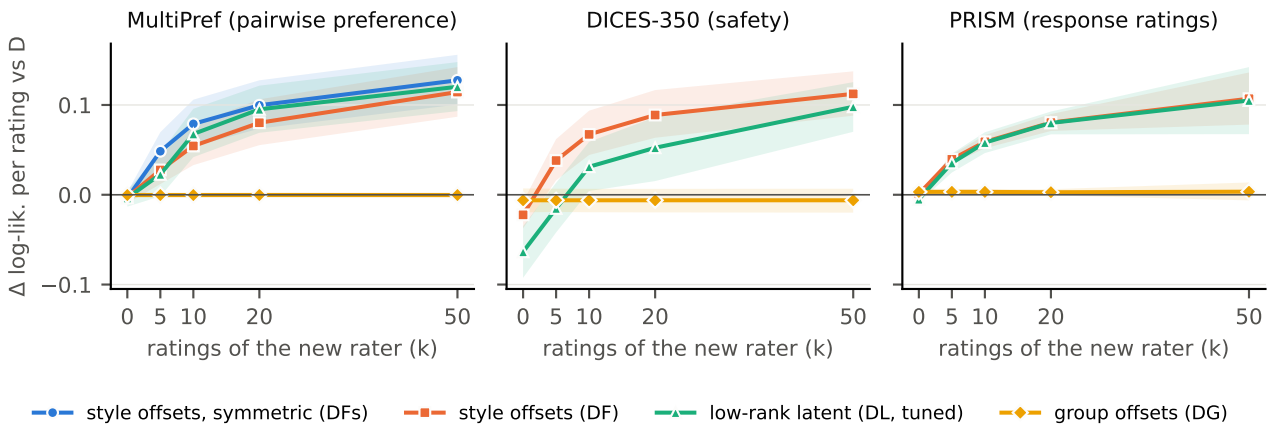


Figure 1. New raters: gain in held-out log-likelihood per rating over the text-only distributional reward model (D), as a function of the number k of the new rater's ratings used for adaptation; 95% rater-cluster bootstrap bands, 5 folds. Text-independent rater offsets (DF; symmetric three-parameter DFs on two-sided preferences) gain 0.08–0.10 nats per rating at k = 20 on all three datasets; group offsets (crowd/expert, race, region) gain nothing; a low-rank text-interacting rater latent with a few-shot-tuned prior matches but does not exceed them (it beats the 5-offset DF on MultiPref and ties DFs). DF, DFs and DG use the training-rater prior; DL the few-shot-tuned prior.

k	DF − D	DFs − D	DG − D	DL − D (tuned prior)	DFs − DL (tuned, post hoc)
5	+0.028 [+0.012, +0.044]	+0.048 [+0.029, +0.070]	≈ 0	+0.023 [−0.002, +0.047]	+0.026 [+0.016, +0.037]
20	+0.080 [+0.055, +0.106]	**+0.100 [+0.074, +0.128]**	≈ 0	+0.095 [+0.069, +0.122]	+0.0045 [−0.0004, +0.0092]
50	+0.114 [+0.087, +0.142]	+0.128 [+0.101, +0.156]	≈ 0	+0.120	+0.007 [+0.002, +0.013]

is prevented (KL weight selected as 0 or 0.001; +0.087 over its own unpersonalised version), but only lifts it to the level of the text-only model; PAL-B's few-shot user weights give no gain. Their user representations are unrelated to response style (cross-validated $R^2 \approx 0$ against the style offsets). On binary labels — the setting in which both methods were proposed — every method's personalisation gain is at most 0.002 nats (small but nonzero for PAL-B with K = 4 and for VPL at k = 50), and VPL's text head is the best binary model (the style model is 0.010 nats behind it and 0.005 behind the text-only model): the individual differences that are predictable for new raters live in the graded levels and ties that binary preference data discard. VPL's per-response reward head is slightly better at predicting *which* response is preferred (0.677 vs 0.668), an effect present without personalisation. The offsets are also a plug-in: added to VPL (fitted on the same k ratings, VPL frozen) they improve it by +0.088 [+0.066, +0.111] nats per rating at k = 20, tie with the style model on likelihood (−0.004 [−0.014, +0.005]) and keep VPL's direction accuracy (+1.1 points over the style model alone).

5.6. Summary of baselines (MultiPref, new raters, k = 20)

Gains are relative to each block's text-only model (D, Dp, D-LoRA); the LoRA block is restricted to the 86 raters of the two in-context folds.

Rows within a block share the text model and the raters; across blocks the text models differ (text-only direction accuracy 0.674 / 0.670 / 0.677). VPL's and PAL-B's own gains over their unpersonalised versions are +0.087 and −0.000.

6. What personalisation buys: calibration, not ranking

Binary labels remove the signal. On MultiPref with ties dropped and labels binarised, every method's personalisation gain is at most 0.002 nats per rating (§5.5), and the scale-use baseline's binary accuracy equals the text-only model's (−0.0002). The individual differences that are predictable for new raters live in the graded levels and in ties — exactly what binary preference data discard, and the format in which most personalised reward models are trained and evaluated. **Ranking moves little.** Text-independent offsets change how strongly a rater is predicted to prefer a response, not which response: on MultiPref the scale-use baseline

dataset	DF − D	DG − D (groups)	DF − DL (tuned prior)
DICES-350 (safety)	+0.089 [+0.064, +0.117]	−0.006 (race)	+0.037 [+0.022, +0.053]
PRISM (ratings)	+0.080 [+0.071, +0.090]	+0.003 (region)	−0.001 [−0.006, +0.003] (formally not non-inferior: bound $-0.006 < -0.005$)

text model	personalisation	gain in LL per rating vs the block's text-only model	direction accura
frozen features (joint item embedding)	group offsets (crowd / expert)	≈ 0	0.677
	low-rank latent, tuned prior	+0.095 [+0.069, +0.122]	0.679
	style offsets, 3 numbers (DFs)	**+0.100 [+0.074, +0.128]**	**0.679**
frozen features (per-response embeddings)	VPL (reimplemented)	+0.002 [−0.008, +0.012]	0.677
	PAL-B (reimplemented)	−0.065 (worse)	0.666
	DFs	**+0.094 [+0.069, +0.120]**	0.668
	VPL + style offsets	+0.090	0.679
LoRA-fine-tuned Qwen2.5-7B (86 raters)	in-context rater history (RC-LoRA)	+0.048 [+0.009, +0.086]	0.647
	DFs	**+0.090**	0.677

raises direction accuracy by only +0.005 [+0.001, +0.008] at k = 20 (and the 5-offset version with an A-vs-B term lowers it by 0.014), while RPS drops from 0.1593 to 0.1532. On single-response safety ratings the same offsets do move the safe/unsafe decision (+4.8 points on DICES), because there a rater's threshold *is* the decision. Content-interacting personalisation does not do better on ranking: VPL's advantage in direction accuracy is present without personalisation (k = 0) and comes from its text head. **Scale use does not corrupt the shared reward model.** Training the text model jointly with rater offsets and then using the population-mean offsets (k = 0) does not give a better consensus model than ordinary training: −0.003 [−0.010, +0.003] nats on MultiPref (direction +0.3 points, n.s.), −0.022 [−0.037, −0.008] on DICES, +0.001 on PRISM, −0.009 [−0.016, −0.004] with the LoRA reward model. Ordinary training averages scale use out; it matters for predicting a person, not for learning the content preference.

7. Recommendations

(i) Collect and keep graded labels (and ties) when individual differences matter; binary preferences erase most of what is predictable about a new rater. (ii) Report a scale-use baseline (a few text-independent offsets per rater) next to any personalised reward model; a method that does not beat it has not shown that it learns more than scale use. (iii) Evaluate on raters unseen in training, adapted from k of their own ratings, and on held-out items. (iv) Report calibration (log-likelihood, RPS) and ranking (direction accuracy) separately: they answer different questions — how strongly someone prefers, and what to show them. (v) Check what a learned user representation encodes; in the two published models we reimplemented it was unrelated to scale use ($R^2 \approx 0$) and carried little else that transferred to new raters.

8. Protocols and what did not hold

All analyses followed timestamped internal protocols (not public registrations); changes after results are logged as amendments with reasons, and the symmetric style model DFs and its hypothesis were added after the first full result. Not supported or not robust: (i) the stage-1 decision rule returned "mixed" (text decodability not ≤ 10 %); (ii) non-inferiority of the 5-parameter DF to the low-rank latent (H3: DF − DL ≥ -0.005) holds with the training-rater prior but not with a few-shot-tuned prior on MultiPref (−0.015) and is formally not met on PRISM (bound −0.006); (iii) "scale beats shift" (H4) holds only on MultiPref; (iv) the originally registered probit-family H5 is not supported (F − D = −0.029 [−0.049, −0.008]; superseded by the distributional family in Amendment 2); (v) the prediction that published user latents encode style (C3) failed ($R^2 \approx 0$); (vi) the hybrid's non-inferiority to DFs (C4, second part) is formally not met (bound −0.014). Four baseline problems found during the study were fixed in favour of the baselines — a missing constant column in the low-rank latent, a scale non-identifiability in the probit group model, VPL's posterior collapse, and PAL-B's near-unregularised adaptation — and the runs before each fix are reported.

9. Limitations

Frozen-feature text heads for the main tables (a fine-tuned head in §5.4); three datasets, one language; PRISM scores binned to 5 levels; rater offsets are text-independent by design — interactions with content exist in principle and are what the low-rank latent can capture; we test linear decoders for the text-side measurement only; adaptation uses the rater's ratings of other items, not of the same item.

A. Protocols and amendments

All analyses were specified in timestamped internal protocols (PROTOCOL_STAGE1, PROTOCOL_STAGE2, PRO-

TOCOL_STAGE2B, PROTOCOL_STAGE2C); stages 2b and 2c were written after the stage-2a results. Changes after a result were logged as numbered amendments with the reason; the main ones: the first decoding split (a = b = 2) was structurally empty (two ratings determine their distribution) and was replaced by a = 3, b = 1; hand-set penalties in the rater–item model over-fitted and were replaced by inner cross-validation (which reversed an earlier, wrong conclusion that rater shifts do not help); penalty grids were extended when selections fell on grid edges, before any hypothesis was read; the low-rank baseline lacked a constant column and was fixed in its favour; the probit group model had a scale non-identifiability and was fixed; VPL suffered posterior collapse and its KL weight was re-selected including 0; PAL-B's adaptation ridge was re-selected by k-shot validation. Every superseded full run is kept and reported in the supplementary results; two debugging smoke runs and one fold whose output was deleted unread when the penalty grids were extended are not.

B. Hypotheses and outcomes

Stage 2a (MultiPref, k = 20): H1 DF $>$ D — holds (+0.080); H5 as first registered (probit F $>$ D) — not supported (-0.029), superseded by H1 in Amendment 2; H2 DF $>$ group — holds; H3 DF $\geq$ low-rank $-$ 0.005 — holds under the training-rater prior, not robust under a few-shot-tuned prior (-0.015); H4 rater scale $>$ rater shift — holds on MultiPref only; H6 symmetric style model gains likelihood without losing direction accuracy — holds and robust. Replications: H1r, H2r hold on DICES and PRISM; H3r holds on DICES (robust) and is a tie on PRISM under the tuned prior. Stage 2b: B1, B2, B3 hold. Stage 2c: C1 holds (graded), C2 holds (binary: $|\text{DFs} - \text{D}|$ accuracy -0.0002; all personalisation gains ≤ 0.002 nats), C3 fails (published latents do not encode style, $R^2 \approx 0$), C4 half-holds (+0.088 over VPL; tie with DFs).

C. Implementation

Text features: last-token final-layer states of Qwen2.5-7B-Instruct under a 5-point judge prompt (MultiPref, PRISM) or a 3-level safety prompt (DICES), label-free PCA to 256 dimensions. Folds: 5 rater folds (stratified by group) $\times$ 5 prompt-group folds; validation: 15 % of training prompt groups. k-shot draws: 5 per k, identical across all models (RC-LoRA: draw 0 only). PRISM item folds are by conversation. Bootstrap: rater clusters, B = 2,000. LoRA: r = 16 on all attention and MLP projections, lr 10^{-4}, one epoch, accumulation 8. VPL / PAL-B: official architectures (latent 512, hidden 512; d_pref 512, mlp2-gelu projectors, angle learner) with an ordered-logistic link for graded labels; VPL's KL weight selected from $\{0.01$ (official), $0.001, 0\}$ and PAL-B's adaptation ridge by k-shot validation (non-official choices made to prevent collapse and over-fitting).

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